

LESSON 9-3 • SESSION 2 • PRACTICE • FORM A

Equations from Situations

Learning target – I can write an equation from a described situation and identify its variables. · 6.EE.C.9

REMEMBER

- Say what each letter means. Then **multiply** the input by the rate for one unit.
- **Add** a starting amount or fee only if the story gives one: \$5 plus \$2 each is $c = 2n + 5$.
- If one person is always 3 years older, **add**: $a = b + 3$.

WORKED EXAMPLE

A pet sitter earns \$9 each day. Write an equation for dollars e earned in d days.

Step 1 Define: d = days, e = dollars earned.

Step 2 Rate: \$9 for each day, so $e = 9d$.

Step 3 Test $d = 5$: $e = 9 \times 5 = 45$.

Answer $e = 9d$; 5 days earns \$45

WORDS unit rate (*tasa unitaria*) · multiplicative rule (*regla multiplicativa*) · additive rule (*regla aditiva*)

PRACTICE

show your work · label units

1. Write an equation for each situation.

- a. \$11 per hour; cost c for h hours: $c = \underline{\hspace{2cm}}$
- b. \$14 per item; cost c for n items: $c = \underline{\hspace{2cm}}$
- c. Mia is 7 years older than her brother. Mia's age m when he is b : $m = \underline{\hspace{2cm}}$
- d. \$20 fee plus \$5 an hour; cost c for h hours: $c = \underline{\hspace{2cm}}$

2. **Paddleboard Rental.** A lake shop rents paddleboards for \$9 each hour.

- a. Name the independent and dependent variables and their letters. _____
- b. Write the equation: $c = \underline{\hspace{2cm}}$
- c. Find the cost for 6 hours. _____ dollars

3. **Lawn Mowing.** Devin earns \$13 for every lawn he mows. Write an equation for his pay p for n lawns. Find his pay for 8 lawns.

Hint The pay repeats for every job, so multiply.

Show your work.

Answer _____ dollars

4. **Big Sister.** Mia is always 7 years older than her brother. Write an equation for Mia's age m when her brother is b . How old is Mia when he is 15?

Hint The age gap stays the same, so add.

Show your work.

Answer _____ years

5. **Field Trip Buses.** Each bus seats 48 riders. Write an equation for the riders r on b buses. Find the riders on 6 buses.

Hint Name what the school chooses, then multiply.

Show your work.

Answer _____ riders

- 6. Two Different Rules.** Graph A passes through (1, 5), (2, 10), (3, 15). Graph B passes through (1, 5), (2, 6), (3, 7). Write an equation for each. How are the rules different?

Hint Is y a multiple of x , or x plus a number?

Show your work.

Equations _____

- 7. Summer Savings.** Ellis saves \$45 from every paycheck. He had nothing saved before.

Hint The starting amount is added once.

- a. Write an equation for his savings s after p paychecks. _____
- b. Suppose he had \$60 saved before. Write the new equation. _____
- c. Use the new equation to find the savings after 4 paychecks. _____ dollars

- 8. Hall Rental.** A hall charges a \$75 booking fee plus \$30 an hour. Write an equation for the cost c after h hours. Find the cost for 5 hours.

Show your work.

Answer _____ dollars

FIND THE ERROR

explain the mistake, then fix it

- 9. Anika's Work.** Buttons cost \$2 each. Anika wrote $c = n + 2$ for the cost c of n buttons. Explain her mistake and fix the equation.

TEST PRACTICE

choose the best answer

- 10.** At \$9 an hour, the cost c after h hours is –
- A** $c = h + 9$ **B** $c = 9h$ **C** $h = 9c$ **D** $c = 9h + 9$
- 11.** A \$20 fee plus \$5 an hour gives –
- A** $c = 20h + 5$ **B** $c = 25h$ **C** $c = 5h + 20$ **D** $c = 5h$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

15 X	105 Q	22 M	240 O	225 L	288 B	104 Y	525 Z	54 S
		#2c	#3	#4	#5	#7c	#8	

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____